

PROGRAMMING FOR ARTIFICIAL INTELLIGENCE

Abdul Haseeb
BS(AI)-IV

Outlines

What is a Variable?

Why use Variables?

How Variables help you in AI Projects?

Types of Variables/Data Types

Variable Naming Rules in Python

Input function

1 What is a Variable?

A **variable** is a name given to a value so we can **store, reuse, and change** it easily.



Think of a variable as a **labeled box** that holds data.

2 Why Use Variables?

Bad Practice (No Variables)

✗ `print("Ali")`
✗ `print("Ali")`
✗ `print("Ali")`

Bad Practice (No Variables)



But if the
name changes?

Good Practice (Using Variable)

✓ `name = "Sara"`
✓ `print("Sara")`
✓ `print("Sara")`
✓ `print("Sara")`

Change Once, Reflected Everywhere

So, why do we use **variables** instead of writing data directly?



Reusability

Write once, use many times.



Readability

Variables make code clear and understandable.



Changeability

Easily update data without changing the code.



Memory Management

Store data efficiently in memory

3 How Variables Help You in AI Projects

AI projects deal with **large, changing data**.
This is where **variables** come in!

Example: Air Quality Prediction (AI)

- ✓ `pm25 = 180`
- ✓ `pm10 = 240`
- ✓ `temperature = 34`

Same Code, New Data

Example: AI Model Training

- ✓ `epochs = 50`
- ✓ `learning_rate = 0.01`
- ✓ `accuracy = 0.92`

This is the foundation of AI & Machine Learning.

Without **variables**, AI is **impossible**!



Reusability

Write once,
use many times.



Readability

Variables make clear
and understandable.



AI models learn

AI models learn.



AI models optimize

Store data efficiently



4 Types of Variables in Python



Python **decides** the type automatically

1 Integer (int)

```
age = 20  
students = 45
```

Used for:

- ◆ counting, loops,
- ◆ indexes

2 Floating Point (float)

```
temperature = 36.5  
accuracy = 0.92
```

Used heavily in:

- ◆ AI, ML, statistics
- ◆

3 String (str)

```
name = "Ali"  
city = "Hyderabad"
```

Used for:

- ◆ labels, messages,
- ◆ categories

Python decides the type dynamically.



Python

...-....

```
x = 10      # int  
x = 10.5    # float  
x = "AI"    # string
```


Variable Naming Rules in Python

A variable name must start with a letter (a–z,A–Z) or an underscore (_)

A variable name cannot start with a number

A variable name can contain letters, numbers, and underscores only

Spaces are not allowed in variable names

Special characters (@, #, \$, %, -, +, etc.) are not allowed

Variable names cannot be Python keywords (if, for, class, etc.)

Variable names are case-sensitive (age and Age are different)

Use snake_case for multi-word variable names (Python convention)

Variable names should be meaningful and descriptive

Input()



To take input from the user in Python, we use `input()` function



Let's dive into details using notebook.